

**GATE-ALL BRANCHES
ENGINEERING MATHEMATICS**



SINGLE VARIABLE CALCULUS

Lecture No.- 9



By- Rahul sir

Recap of Previous Lecture



Topic

Concepts based on definite integrals



[Imp ✓ Area Bounded regions
✓ Improper Integrals]

[After
Linear
Algebra]

Topics to be Covered



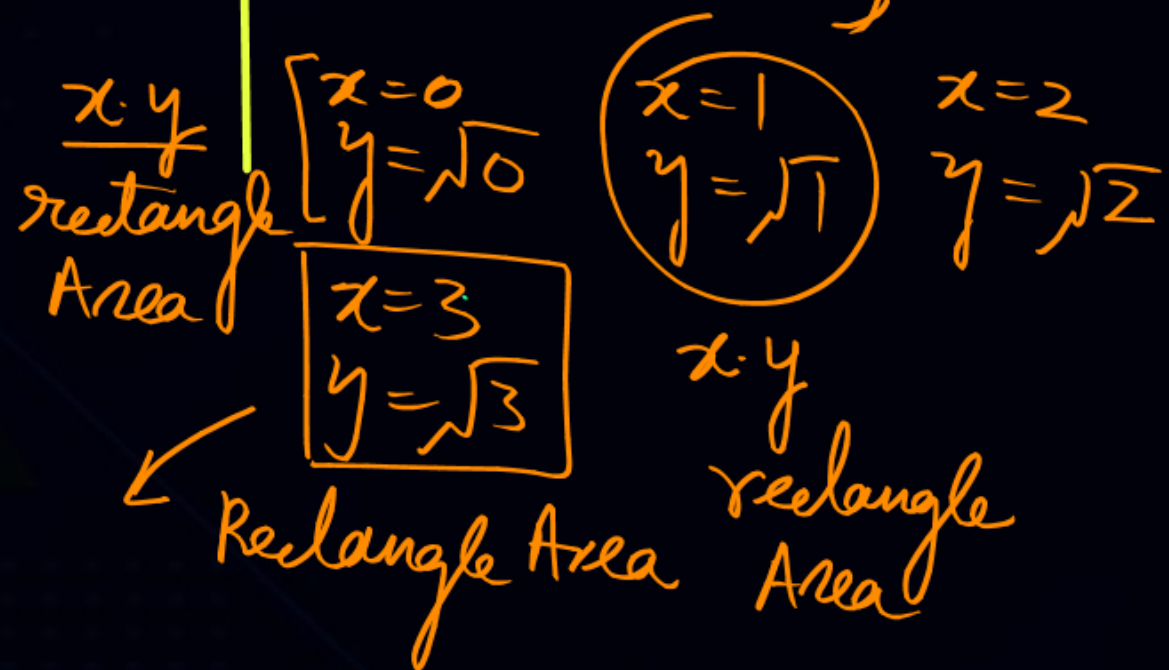
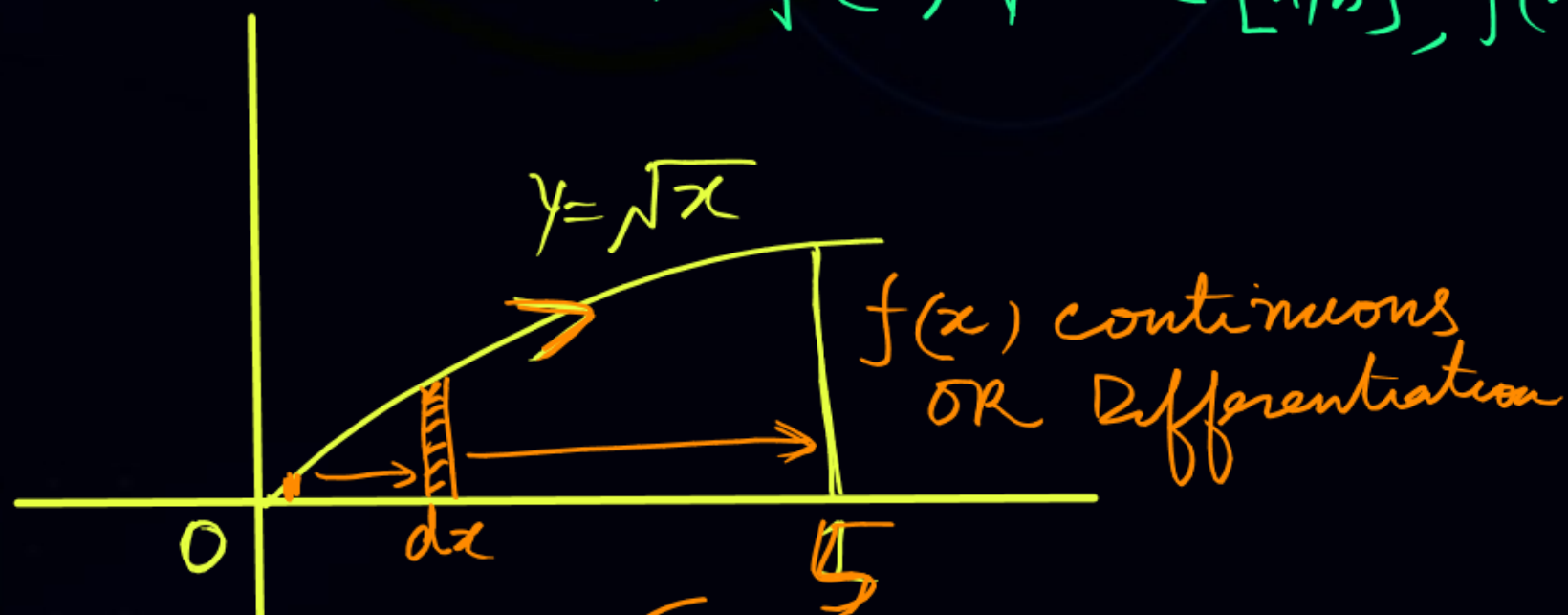
Topic

Problems based on definite integrals

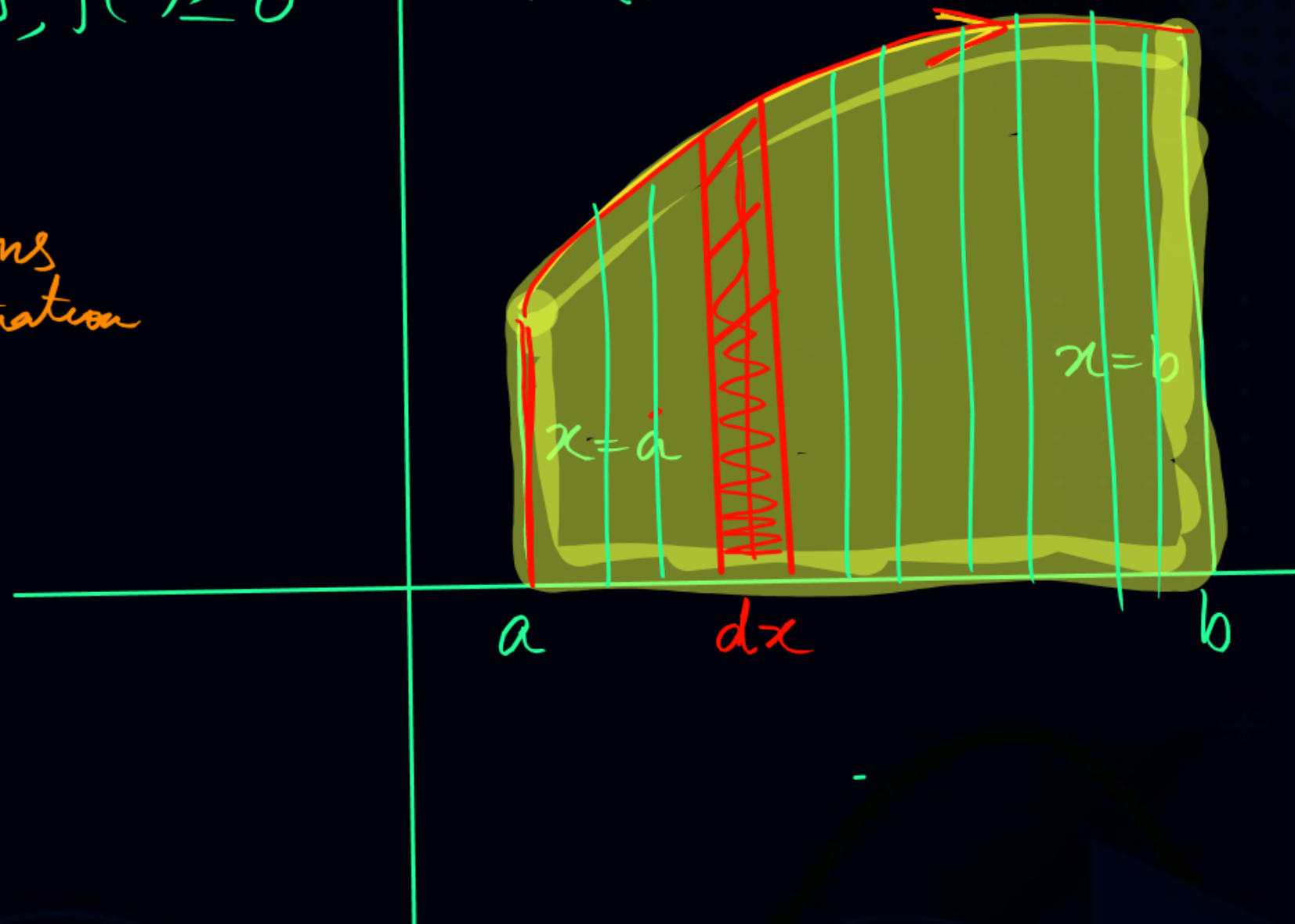


Area Bounded regions:

$$y = f(x) \quad \forall x \in [a, b], \quad f(x) \geq 0$$



Above The x-axis $y = f(x)$

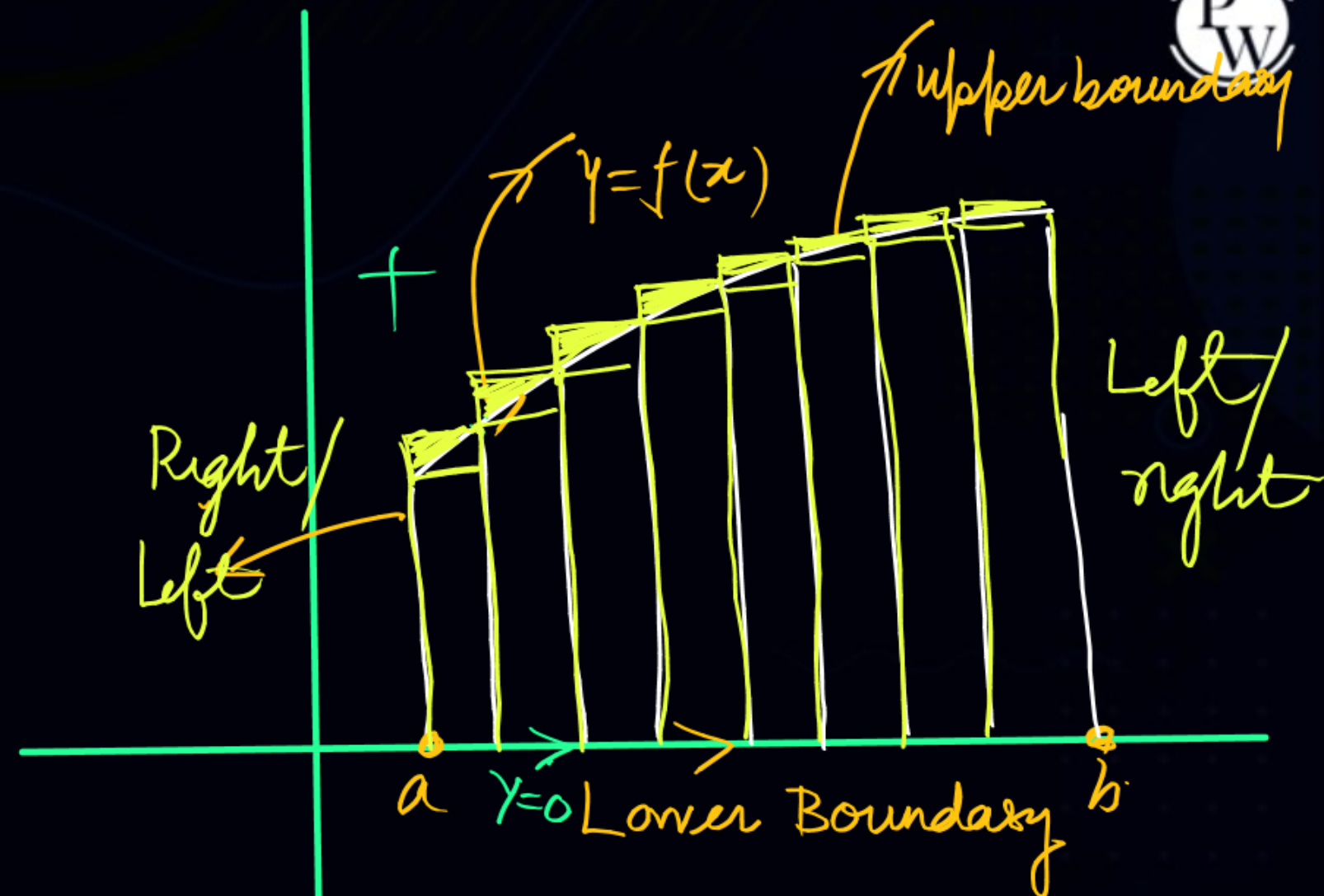


Area Bounded by a curve OR
Area Under A Curve

$$A = \int_a^b [f(x) - 0] dx$$

$$\text{Area} = \int_{\text{Left}}^{\text{Right}} (\text{Upper} - \text{Lower}) dx$$

Area above The X-axis = $\int_a^b y dx$



All rectangle sum = $\sum \oplus \rightarrow \int$

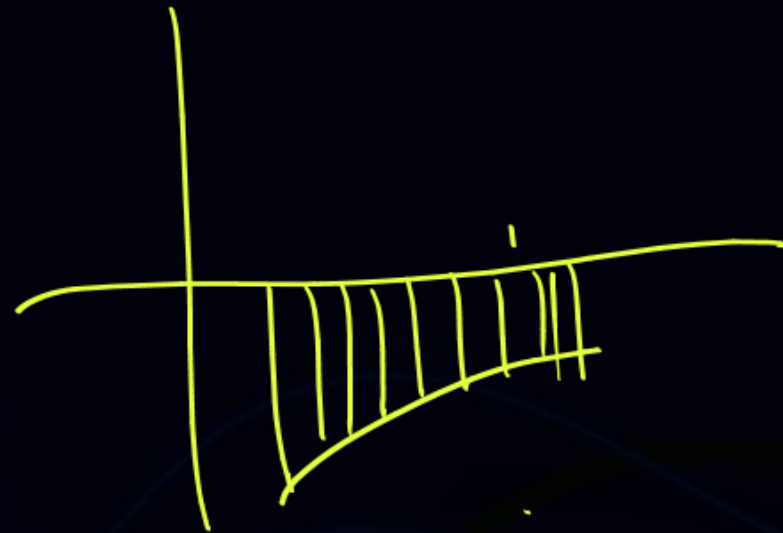
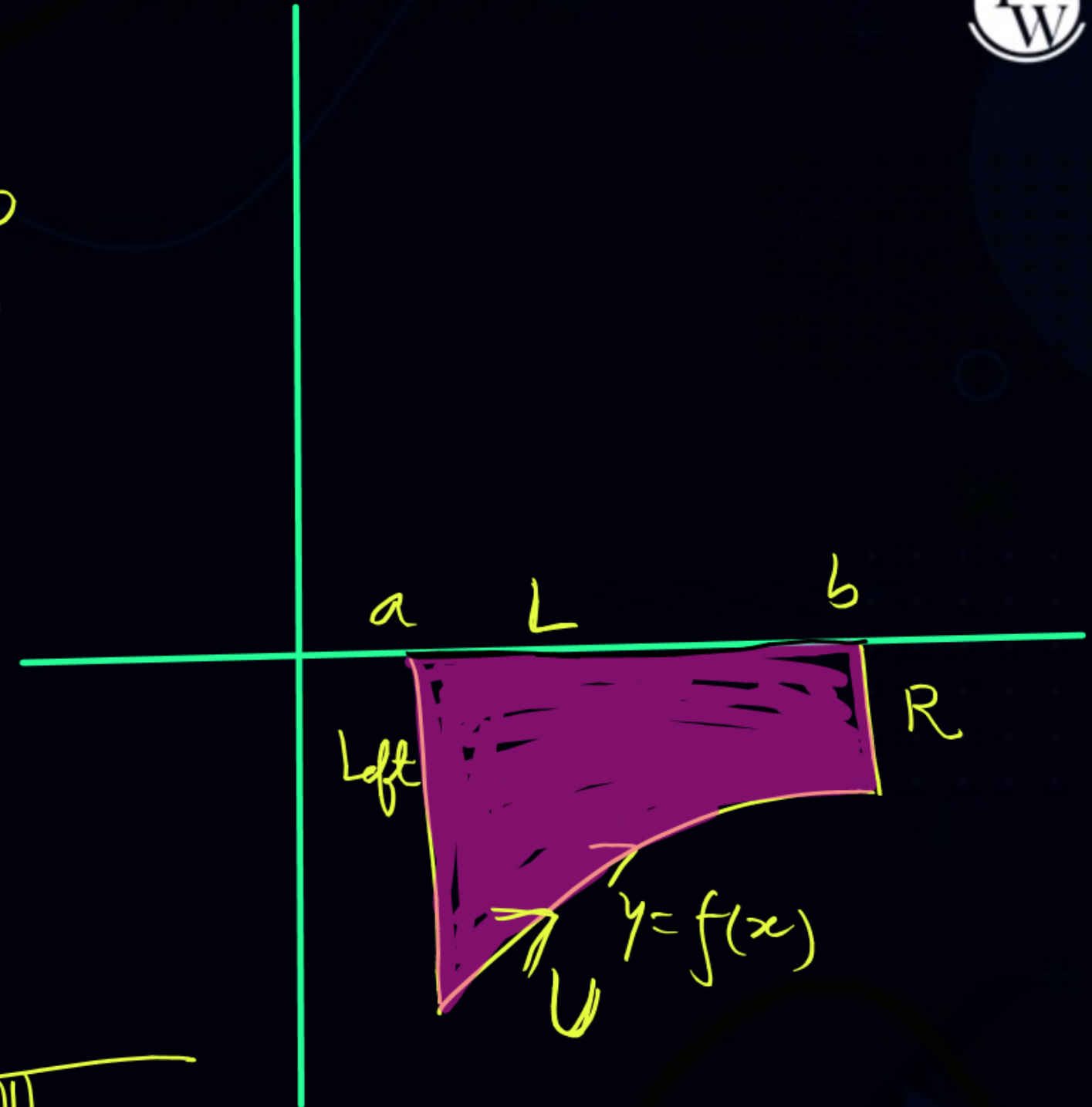
CASE 02 :

$$y = f(x) \quad \forall x \in [a, b] \quad f(x) \leq 0$$

(below the x-axis)

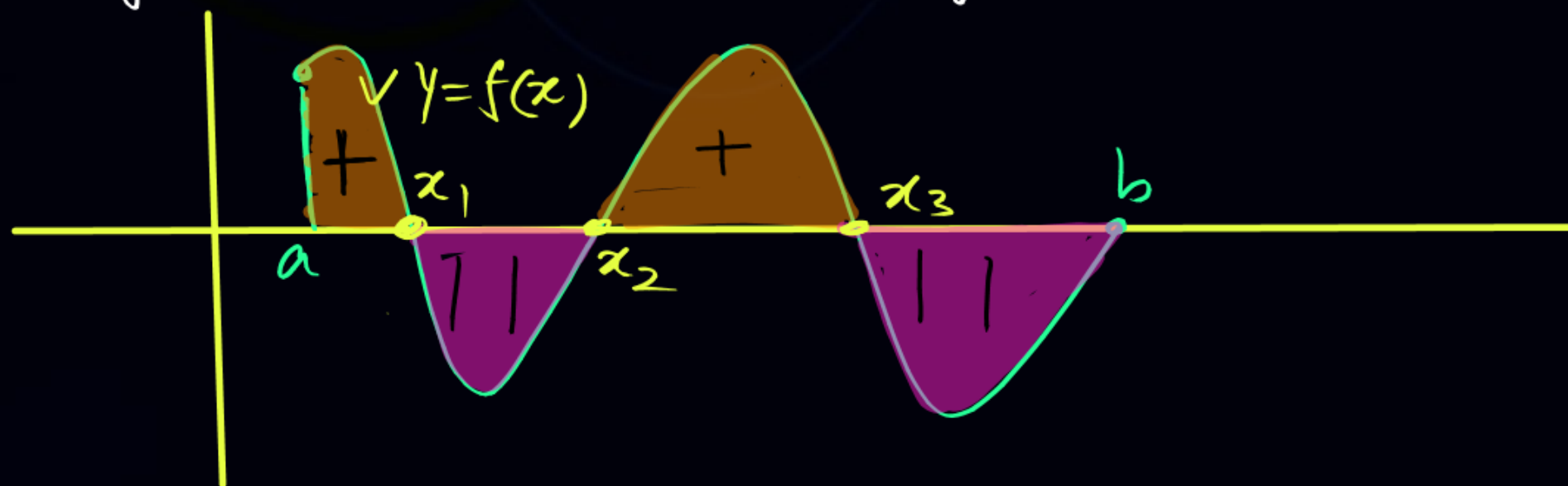
Area under A curve

$$A = \left| \int_a^b f(x) dx \right|$$



CASE 03 :

If $y = f(x)$ crosses many Points.



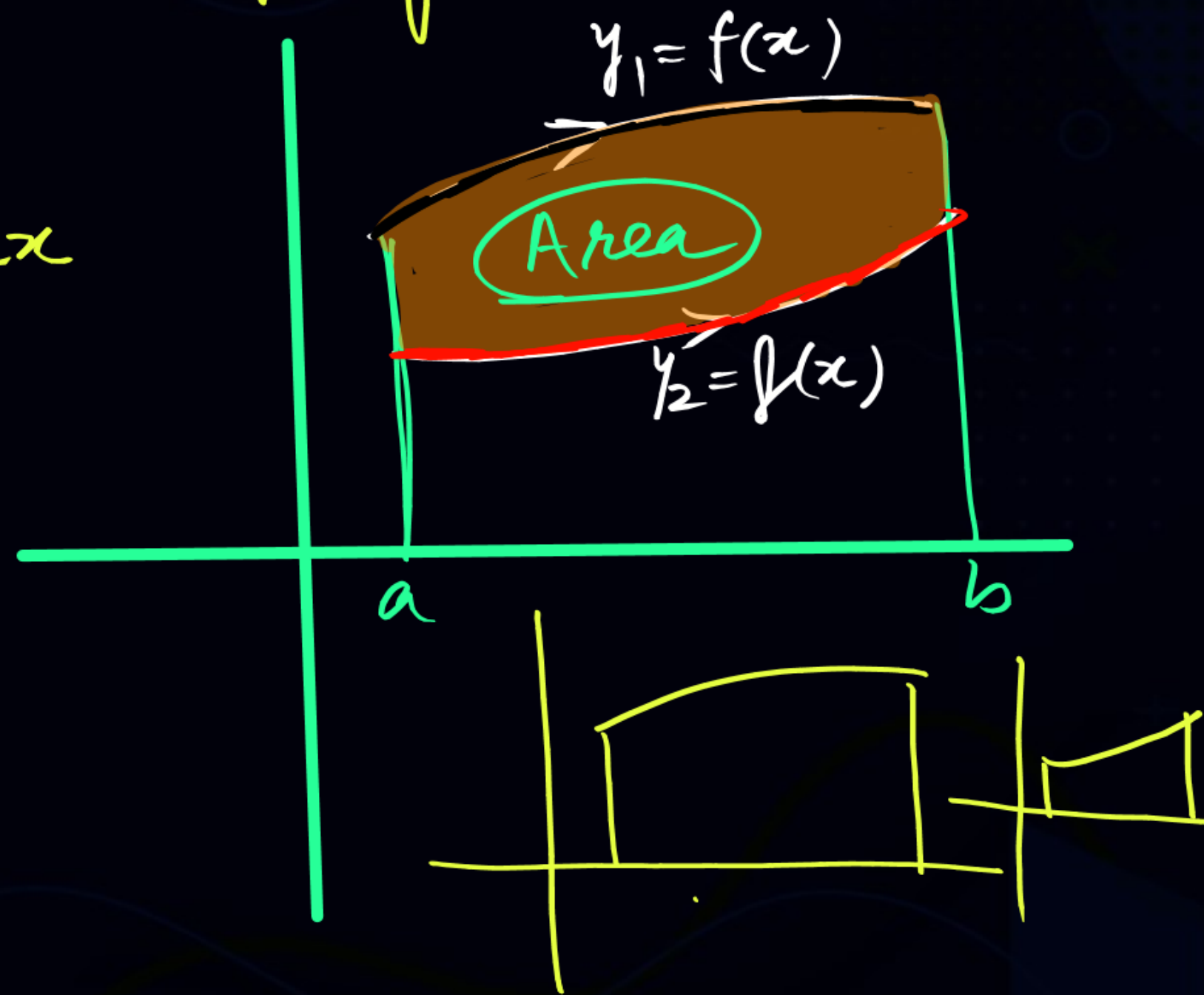
$$\int_a^b f(x) dx = \underbrace{\int_a^{x_1} f(x) dx}_{\substack{\text{Above} \\ \text{The } x\text{-axis}}} + \underbrace{\left| \int_{x_1}^{x_2} f(x) dx \right|}_{\substack{\text{below} \\ \text{The } x\text{-axis}}} + \underbrace{\int_{x_2}^{x_3} f(x) dx}_{\substack{x\text{-axis} \\ \text{above}}} + \underbrace{\left| \int_{x_3}^b f(x) dx \right|}_{\substack{\text{below} \\ x\text{-axis}}}$$

CASE 04If $y_1 = f(x)$ $y_2 = g(x)$

Area Bounded Region

$$\text{Area} = \int_a^b [f(x) - g(x)] dx$$

$$\text{Area} = \int_a^b (y_1 - y_2) dx$$



find The Area Bounded regions

Step 1 $\Rightarrow$ Plot The curve

Step 02: find out The Boundary Point

$$2y = x^2, y = x + 2$$

$$2(x+2) = x^2$$

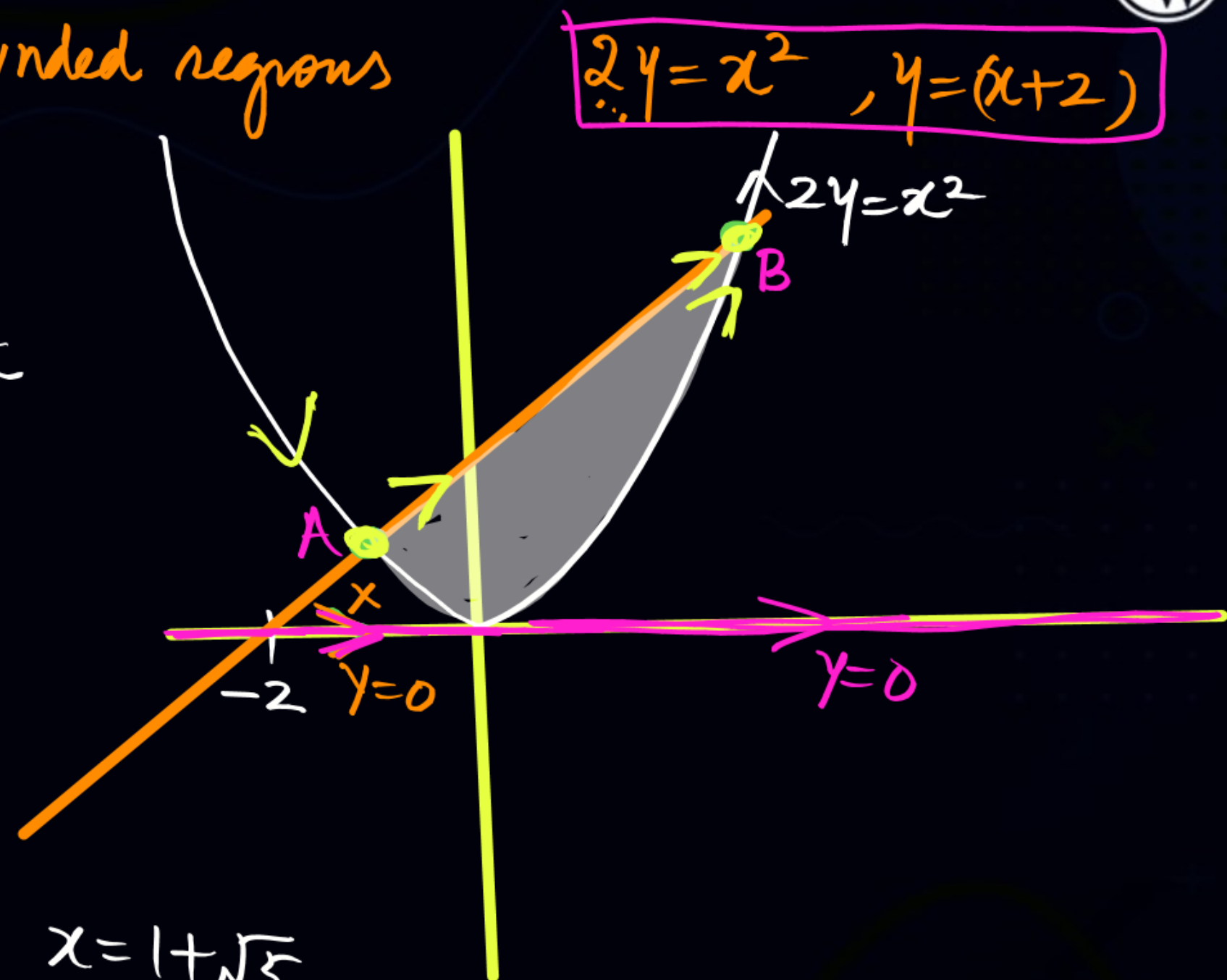
$$2x + 4 = x^2$$

$$\Rightarrow x^2 - 2x - 4 = 0$$

$$\Rightarrow x = \frac{+2 \pm \sqrt{4 + 4 \times 4 \times 1}}{2}$$

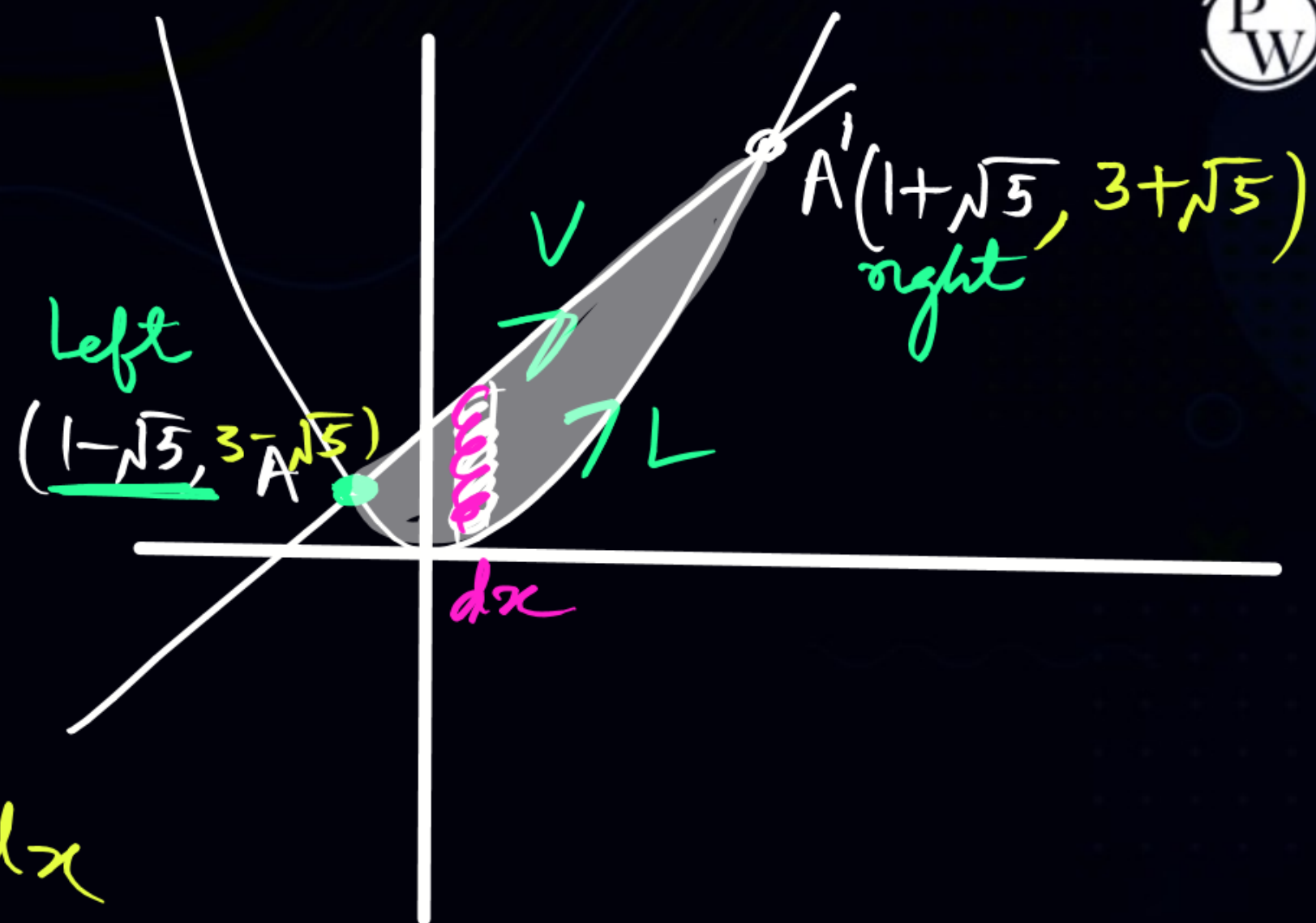
$$x = \frac{2 \pm \sqrt{20}}{2} = \underline{1 \pm \sqrt{5}}$$

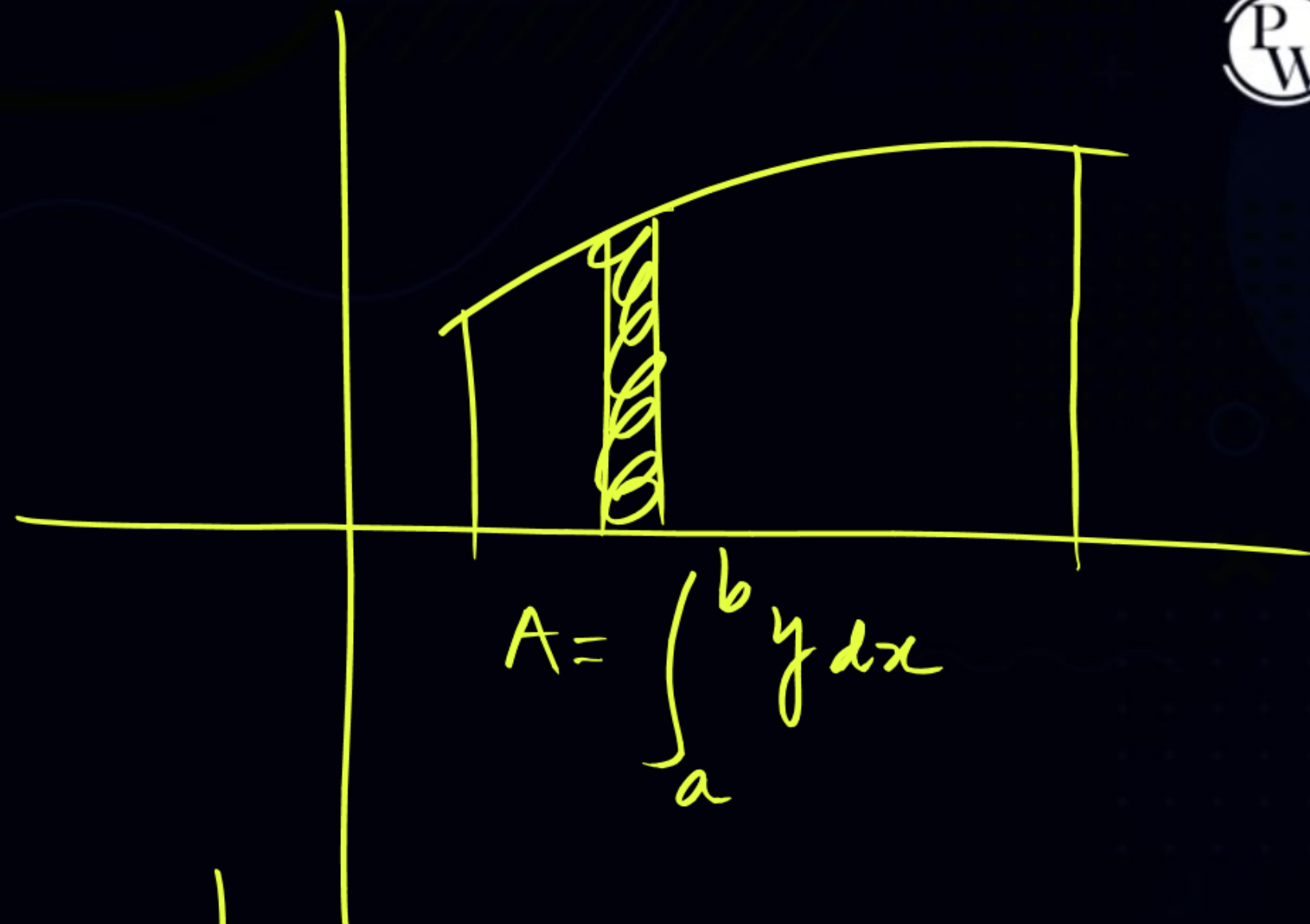
$$\begin{aligned} x &= 1 + \sqrt{5} \\ x &= 1 - \sqrt{5} \end{aligned}$$



$$\begin{aligned}
 y &= x+2 \\
 x &= 1+\sqrt{5} \\
 y &= 3+\sqrt{5} \\
 x &= 1-\sqrt{5} \\
 y &= 1-\sqrt{5}+2 \\
 &= 3-\sqrt{5}
 \end{aligned}$$

$$\Rightarrow \int_{1-\sqrt{5}}^{1+\sqrt{5}} \left[(x+2) - \left(\frac{x^2}{2} \right) \right] dx = \underline{\text{Ans}}$$





$$\text{Area} = \int_{y=a}^b x \, dy$$





Topic : Definite Integrals



#Q. Let the straight line $x = b$ divide the area enclosed by $y = (1 - x)^2$, $y = 0$ and $x = 1$ into two parts R_1 ($0 \leq x \leq b$) and R_2 ($b \leq x \leq 1$) such that $R_1 - R_2 = \frac{1}{4}$. Then b equals

$$y = (1 - x)^2$$

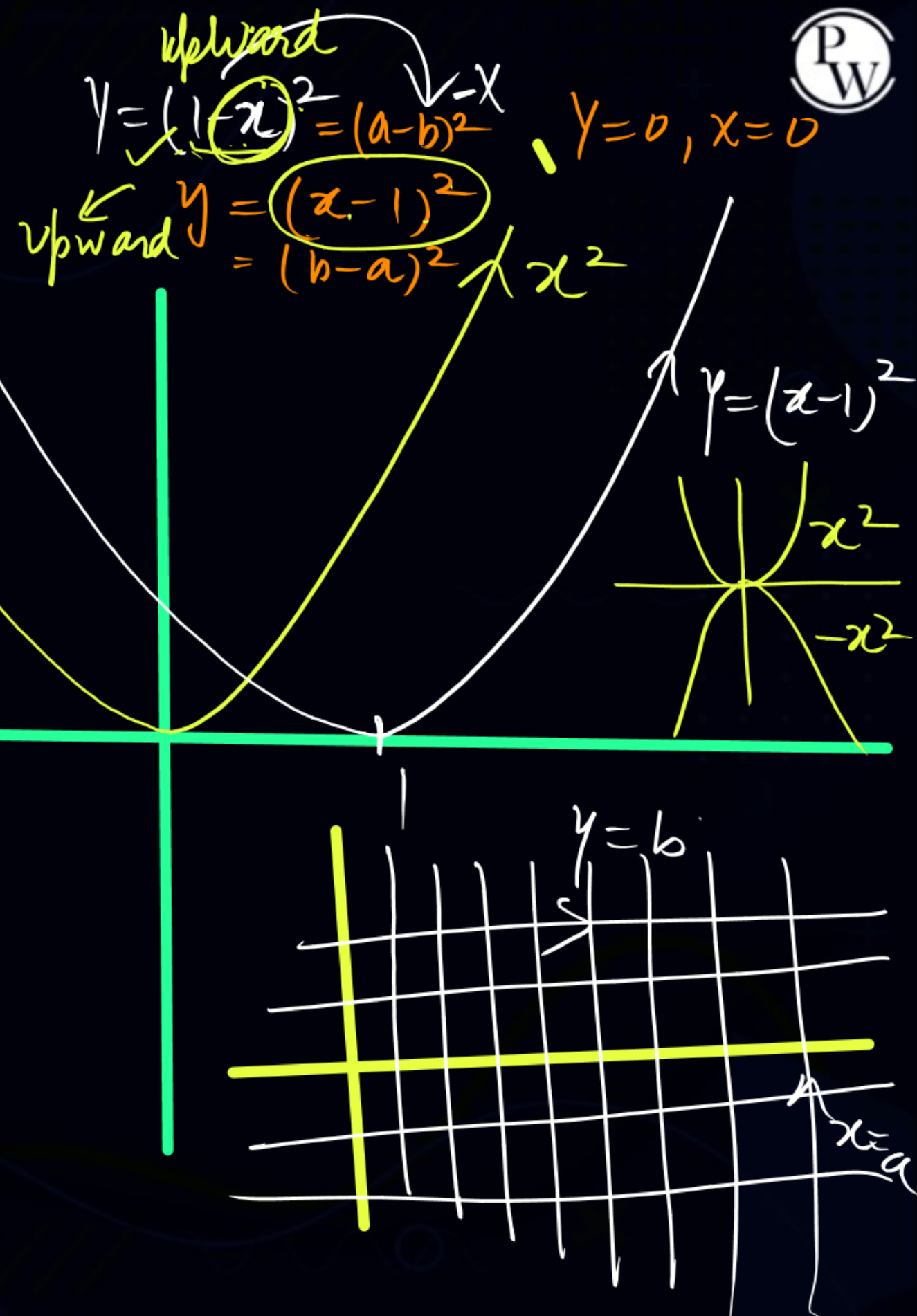
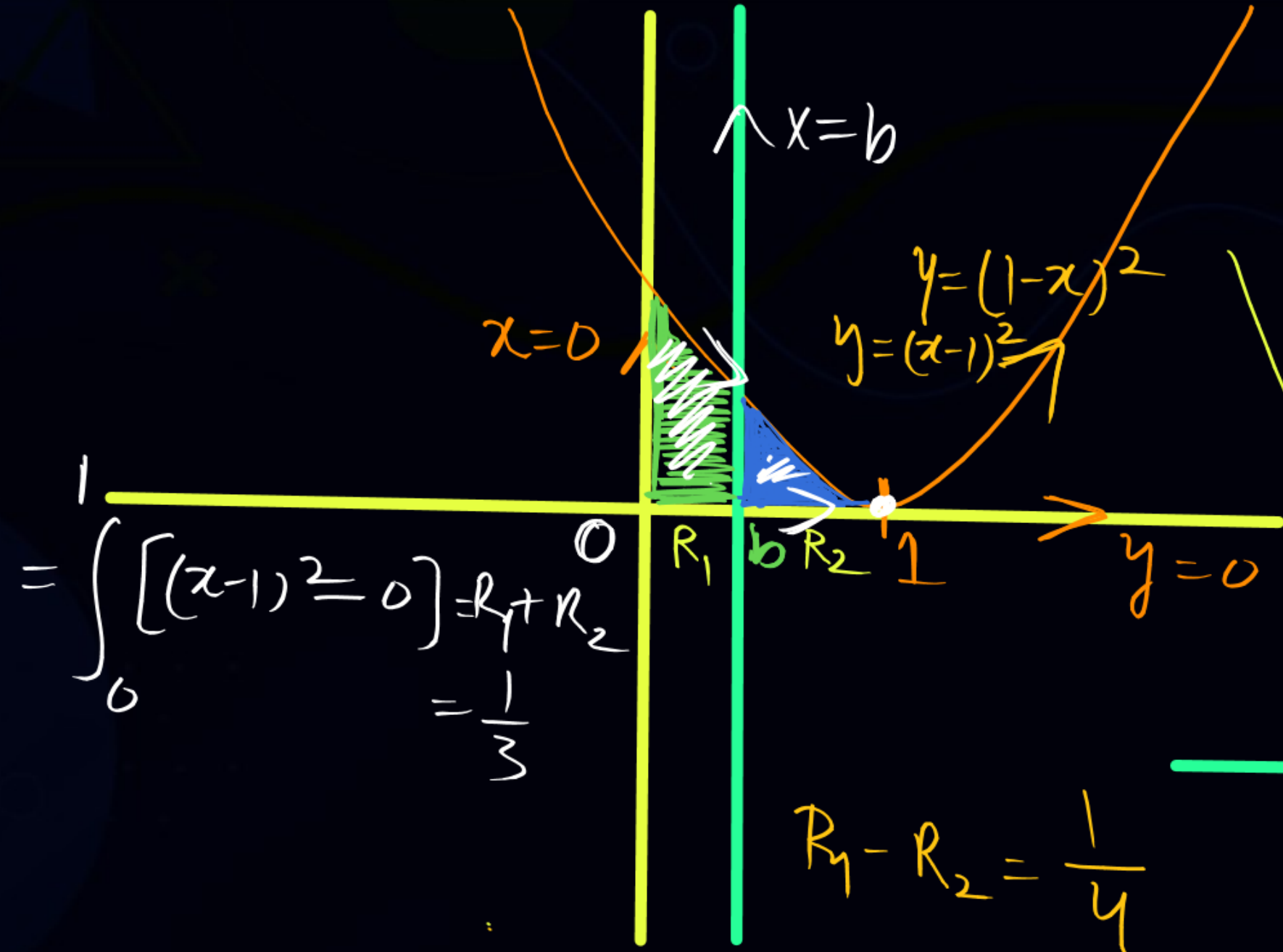
$$y = (1 - x)^2 \quad y = 0 \quad x = 1$$
$$R_1 (0 \leq x \leq b) \quad R_2 (b \leq x \leq 1)$$

A $\frac{3}{4}$

B $\frac{1}{2}$

C $\frac{1}{3}$

D $\frac{1}{4}$



$$R_1 - R_2 = \frac{1}{4}$$

$$= \int_0^b [(x-1)^2 - 0] dx - \int_b^1 [(x-1)^2 - 0] dx = \frac{1}{4}$$

$$b = \frac{1}{2} \checkmark$$

IInd

$$R_1 + R_2 = \int_0^1 [(x-1)^2 - 0] dx = \frac{1}{3}$$

$$R_1 + R_2 = \frac{1}{3}$$

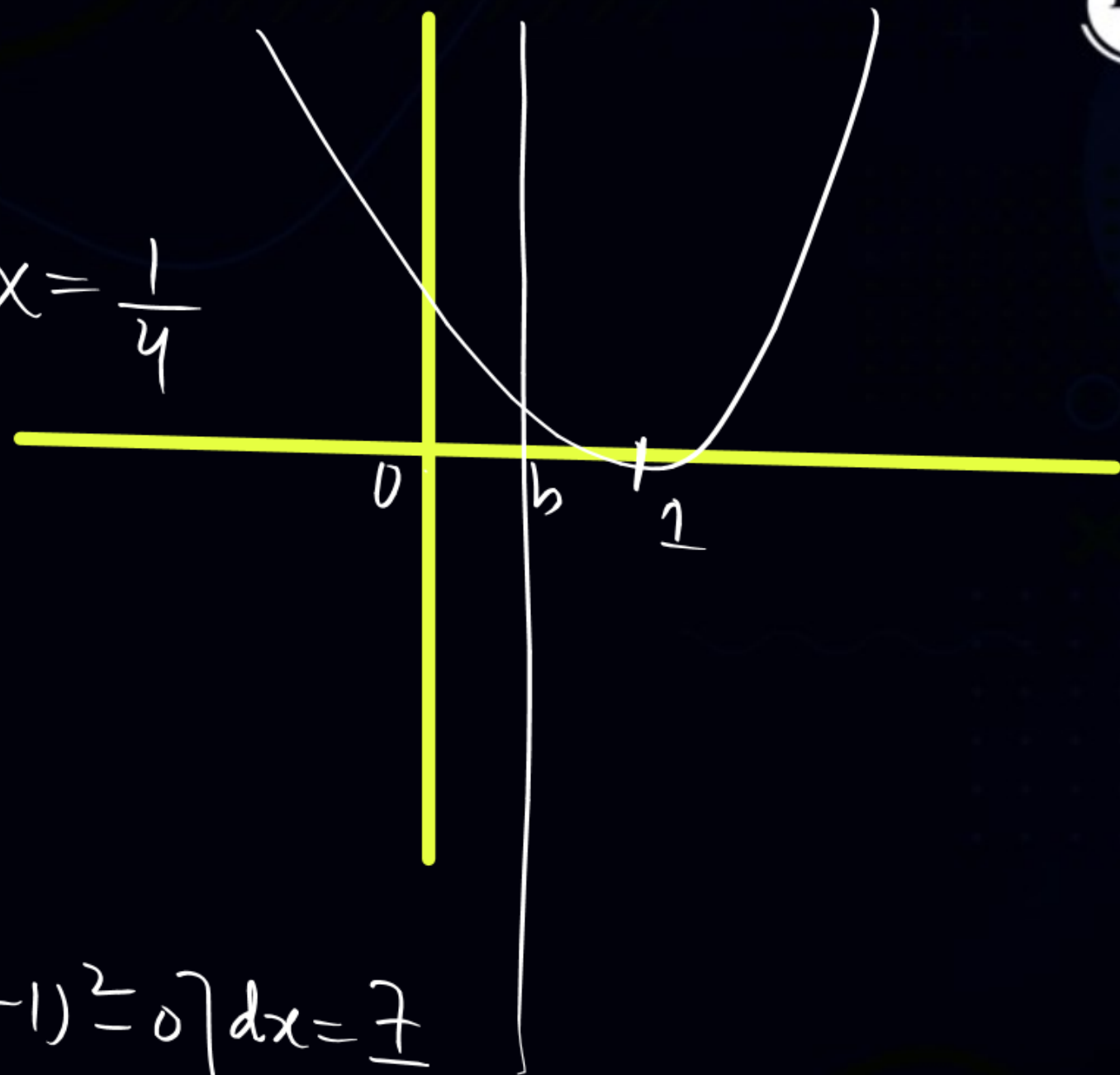
$$R_1 - R_2 = \frac{1}{4}$$

$$2R_1 = \frac{7}{12}$$

$$R_1 = \frac{7}{24}$$

$$\int_0^b [(x-1)^2 - 0] dx = \frac{7}{24}$$

$$b = \frac{1}{2}$$





Topic : Definite Integrals



$$x^2 + y^2 = a^2$$

→ circle radius = $\sqrt{2}$

#Q. The area bounded by the curve $y = x^2$ and $x^2 + y^2 = 2$ above X-axis is

Parabola

A

$$\frac{1}{6} + \frac{\pi}{4}$$

B

$$\frac{\pi}{4} + \frac{1}{6}$$

C

$$\frac{2}{3} + \frac{\pi}{2}$$

D

$$\frac{1}{3} + \frac{\pi}{2}$$

28 October

$$x^2 + y^2 = 2$$

$$y = x^2$$

$$\Rightarrow x^2 + x^4 = 2$$

$$\Rightarrow x^4 + x^2 - 2 = 0$$

biquadratic eqn

$$x^2 = t$$

$$\Rightarrow t^2 + t - 2 = 0$$

$$\Rightarrow t^2 + 2t - t - 2 = 0$$

$$\Rightarrow t(t+2) - 1(t+2) = 0$$

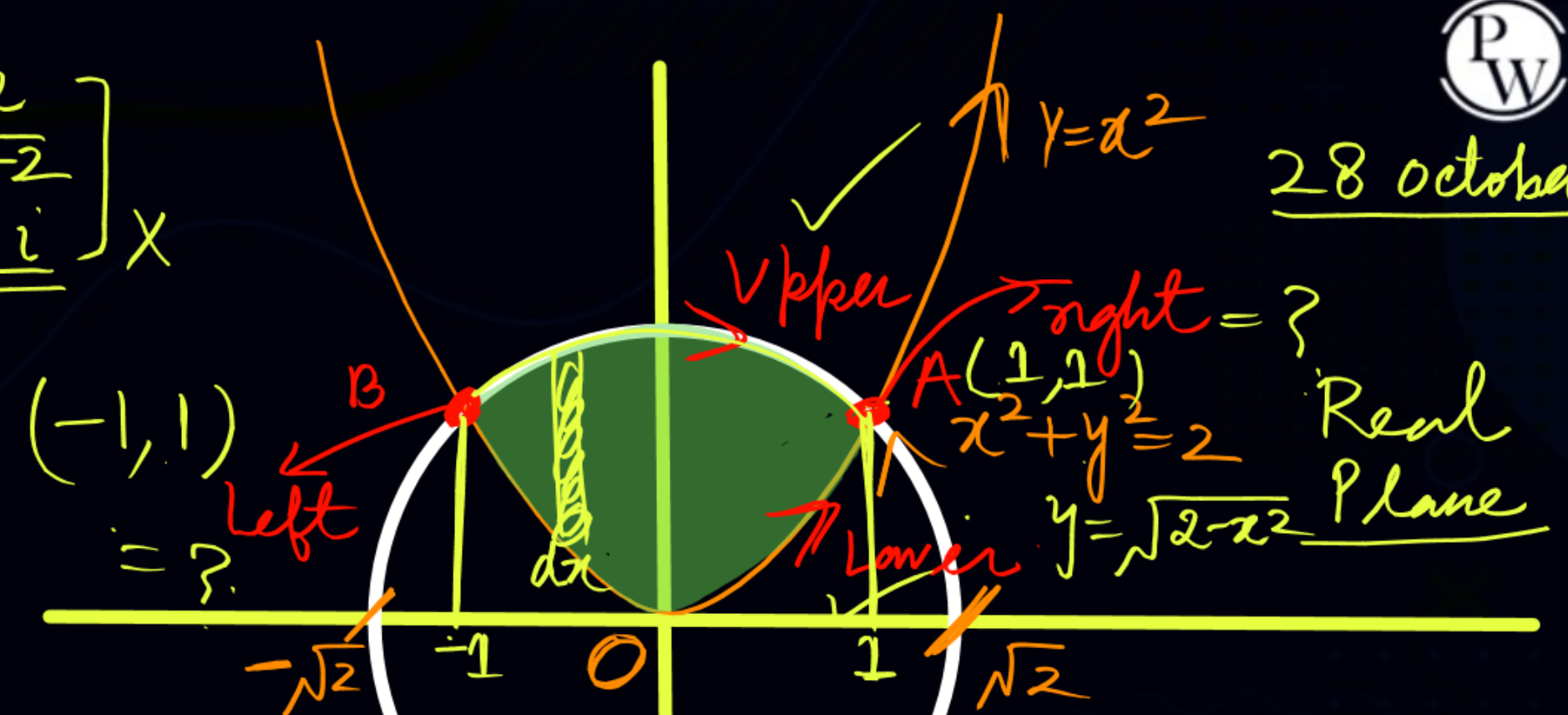
$$\Rightarrow t = 1 \quad t = -2$$

$$x^2 = 1 \quad x^2 = -2 \quad \text{complex}$$

$$x = \pm 1$$

Accepted

$$\left. \begin{aligned} x^2 &= -2 \\ x &= \sqrt{-2} \\ &= \underline{\underline{2i}} \end{aligned} \right\} x$$



$$x = 1$$

$$y = x^2 = (1)^2 = 1$$

$$x = -1$$

$$y = (-1)^2 = 1$$

Area under a Curve

$$= \int_{-1}^1 \left[\sqrt{2-x^2} - x^2 \right] dx$$

$$\int \sqrt{2-x^2} = \int \sqrt{a^2-x^2} dx$$

$$= \frac{1}{3} + \frac{\pi}{2}$$

$$\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right)$$

$$\int \sqrt{2 - x^2} = \frac{x}{2} \sqrt{2 - x^2} + \frac{(\sqrt{2})^2}{2} \sin^{-1} \left(\frac{x}{\sqrt{2}} \right)$$

THANK - YOU